

Compute Moran Eigenvector Maps for Spatial Filtering

Description

Computes Moran Eigenvector Maps (MEMs) from projected core locations using the shared exact or low-rank strategy and returns the public predictor data frame used by `[prepare_spatial_predictors_for_fit()]`.

Usage

```
compute_spatial_mev(
  data_coords_projected = NULL,
  n_mev = 20L,
  strategy = "exact",
  exact_max_locations = 1999L,
  fast_eigenvectors = 200L,
  fast_seed = 900723L
)
```

Arguments

<code>data_coords_projected</code>	A data frame with 'dataset_name' as row names and columns 'coord_x_km' and 'coord_y_km', as returned by 'project_coords_to_metric()'. Must have more than 3 rows (required by 'sjSDM::generateSpatialEV()'). Each row represents one core/site location.
<code>n_mev</code>	A positive integer giving the number of eigenvectors to return. If 'n_mev' exceeds the available count, it is clamped down with a warning. Default is '20L'.
<code>strategy</code>	Shared construction strategy: "exact", "fast", or "auto". Automatic selection uses the shared location boundary.
<code>exact_max_locations</code>	Largest input allowed to use dense exact construction. Default is '1999L'.
<code>fast_eigenvectors</code>	Low-rank basis size reserved for fast construction. It must be at least 'n_mev'. Default is '200L'.

<code>fast_seed</code>	Positive integer used locally for deterministic fast-basis knot selection. Default is '900723L'.
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Details

MEMs capture the spatial autocorrelation structure of the sampling locations and are used as spatial predictors in the sjSDM model. Eigenvectors are computed on the unique core locations; the caller is responsible for expanding to sample level via `'prepare_spatial_predictors_for_fit()'`.

Exact construction uses `'sjSDM::generateSpatialEV()'`. Fast construction uses the package-backed Nyström basis from `'spmoran::meigen_f()'`. This compatibility wrapper returns only the public values; callers that need held-out projection state or provenance should use `[compute_spatial_mev_basis()]`.

Value

A data frame with the same row names as `'data_coords_projected'`, and `'n_mev'` columns named `'mev_1'`, `'mev_2'`, ..., `'mev_n_mev'`, containing the first `'n_mev'` Moran eigenvectors. This data frame is a drop-in replacement for `'dplyr::select(data_coords_projected, coord_x_km, coord_y_km)'` as input to `'prepare_spatial_predictors_for_fit()'`.

See Also

`[project_coords_to_metric()]`, `[prepare_spatial_predictors_for_fit()]`, `[scale_spatial_for_fit()]`